

VR-Numbers.

Operational Superstructures of Integers, Rationals, Reals, and Complex Numbers over the Natural Numbers of VR

Vitaly Reznik

ORCID: 0009-0002-4103-6387

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Abstract

Building on VR (Reznik, 2026), the axiom-free reconstruction of arithmetic from three primitive operations $\{\emptyset, \rightarrow, t\}$, we develop **VR-Numbers** — a program of operational superstructures yielding $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ over VR’s natural numbers $\mathbb{N}$. The construction uses no ordered pairs as primitive objects; each numerical extension is a formal language of expressions with equivalence relations and operations. The entire system rests on the base operation $\emptyset$ alone — a nullary operation, carrying no ontology; all numerical objects — including $\mathbb{N}$ — arise as operational constructions (terms over the operations) of progressively greater depth. Classical actual infinity is replaced by operational infinity: the induction principle is read as an operational principle, not as a postulate of completed infinite totalities. The two-dimensionality of $\mathbb{C}$ is structurally grounded in the duality present in the generativity principle A1 of VR (the two generating implications $F \rightarrow F$ and $F \rightarrow \top$), rather than postulated through pairs of reals; the algebraic coupling of the two axes requires an additional joining rule ($i^2 = \ominus 1$). The structures $\mathbb{Z}_{VR}$, $\mathbb{Q}_{VR}$, $\mathbb{C}_{VR}$ are isomorphic to their standard counterparts; $\mathbb{R}_{VR}$, built from Cauchy sequences with a *computable modulus of convergence*, is isomorphic to the field of computable real numbers — a countable subfield of classical $\mathbb{R}$ — reflecting the operational character of the construction.

Keywords: foundations of mathematics, operational extensions, operational infinity, integer/rational/real/complex construction, Cauchy sequences with computable modulus, axiomatic duality, computable analysis.

Version note (1.1.0). This version synchronises VR-Numbers with the parent work VR v1.1.0 (*An Axiom-Free Reconstruction of Arithmetic from Operations Alone*): Part I now recalls VR with no axioms of its own — the named principles A1–A4 are definitions or theorems over the free term-algebra, membership $\in$ is defined, F and $\emptyset$ are two registers (not one entity). It also (i) removes a pre-1.0.3 ontology relic from the table of §VI.2; (ii) narrows the equivalence claim of §VI.3 to *arithmetical* theorems (via $VR \leftrightarrow$ first-order PA); (iii) makes the computable modulus of convergence explicit in §IV.2 and the apartness of the divisor explicit in §IV.5, closing the Specker gap that secures the isomorphism of §IV.7; (iv) corrects “definable” to “computable (algorithmically approximable)” in §IV.7; and (v) makes the order on $\mathbb{Z}_{VR}/\mathbb{Q}_{VR}$ explicit. No construction, equivalence, operation, or theorem of Parts II–VIII is altered. (Versions 1.0.2–1.0.3 added Part VIII and the no-ontology presentation.)

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Part I. Foundation: VR and $\mathbb{N}$ (axiom-free)

This work builds on the formal system VR (Reznik, 2026, *An Axiom-Free Reconstruction of Arithmetic from Operations Alone*; DOI 10.5281/zenodo.20633314). We recall its essentials in the form fixed by VR v1.1.0.

Primitive operations. VR is generated by three operations: the nullary base $\emptyset$, the unary succession t , and the binary implication $\rightarrow$. Every object is a *term* over this signature; nothing is posited beyond what these operations generate.

No axioms of its own. What earlier versions called the four axioms A1–A4 are, under the operational reading, definitions or theorems over the freely generated term-algebra:

- **A2** defines implication $\rightarrow$ on $\{\mathbf{F}, \top\}$ by the classical truth table.
- **A1** collects two values of that table ($\mathbf{F} \rightarrow \mathbf{F} = \top$, $\mathbf{F} \rightarrow \top = \top$, with $\top := \mathbf{F} \rightarrow \mathbf{F}$) together with a *closure theorem* (the closure of $\{\mathbf{F}\}$ under $\rightarrow$ is all of $\{\mathbf{F}, \top\}$). Here $\mathbf{F}$ (the logical base value) and $\emptyset$ (the operational base object) are *parallel nullary bases in two registers* — written in correspondence, not identified as one entity.
- **A3** is the *theorem* that $t(x)$ has the von Neumann content $x \in t(x)$ and $x \subseteq t(x)$; it follows from the definition of membership below. (t is the primitive succession; $t(x) = x \cup \{x\}$ is a gloss, with $\cup, \{\cdot\}$ abbreviations, not primitives.)
- **A4** (induction) is the *recursor* of the inductive term-algebra: the O_n exhaust the objects generated from $\emptyset$ by t .

Thus VR is axiom-free relative to the type-theoretic framework (the free term-algebra on $\{\emptyset, t\}$ and its recursor); what Peano postulates, VR derives.

Membership is defined. $\in$ is not a primitive but a relation defined by structural recursion: $z \in \emptyset := \mathbf{F}$ and $z \in t(y) := (z \doteq y) \vee (z \in y)$, where $\doteq$ is structural term-identity (coinciding with Leibnizian equality below by theorem).

Natural numbers and equality. $O_0 = \emptyset$, $O_1 = t(\emptyset) = \{\emptyset\}$, $O_2 = \{\emptyset, \{\emptyset\}\}, \dots$; the natural number n is the term O_n . Equality is Leibnizian, $x = y :\Leftrightarrow \forall p : p(x) \leftrightarrow p(y)$, read schematically. VR is arithmetically equivalent to first-order PA and consistent relative to ZF (a metatheoretic corollary).

Operational status. There is no ontological primitive: $\emptyset$ is itself a nullary operation, characterised solely by $\forall x \neg(x \in \emptyset)$. All other objects, the naturals included, are terms over the operations: O_n is not an independent existing thing but the result of n applications of t to $\emptyset$. The content of n is its operational history. This operational position is the foundation of the present work: nothing simply “exists” — objects are terms over operations, of progressively greater depth.

Notation. Expression builders: $a \ominus b$ (integer), $a \oslash b$ (fraction), $a \oplus b \cdot i$ (complex). Operations are written with the symbols $\oplus, \otimes, \boxminus$ (integers), $\boxplus, \boxtimes, \boxminus_{\mathbb{Q}}, \oslash_{\mathbb{Q}}$ (rationals), and a subscript naming the system where needed ($\oplus_{\mathbb{R}}, \otimes_{\mathbb{R}}, \dots$); per §VI.3 every such symbol is read operationally.

Part II. Integers $\mathbb{Z}$ as an operational superstructure

II.1 Language of subtraction. Extend VR by a binary symbol $\ominus$. An expression $a \ominus b$ ($a, b \in \mathbb{N}$) records the operation “subtract b from a ”. It is not an ordered pair but a formal record of an operation.

II.2 Equivalence. $a \ominus b \approx c \ominus d \Leftrightarrow a + d = b + c$ (using only $\mathbb{N}$ -addition). Reflexive, symmetric, transitive (by commutativity/associativity of $+$, and the right-cancellation law $a + c = b + c \Rightarrow a = b$ — see §VIII.1).

II.3 Operations. $(a \ominus b) \oplus (c \ominus d) := (a + c) \ominus (b + d)$; $(a \ominus b) \otimes (c \ominus d) := (a \times c + b \times d) \ominus (a \times d + b \times c)$; $\ominus(a \ominus b) := b \ominus a$; $(a \ominus b) \boxminus (c \ominus d) := (a + d) \ominus (b + c)$ (definable as $\oplus \ominus$). All well-defined on classes.

II.4 Canonical form. If $a \geq b$: $(a - b) \ominus 0$; if $a < b$: $0 \ominus (b - a)$.

II.5 Order. $[a \ominus b]$ is *positive* iff $a > b$ in $\mathbb{N}$ (i.e. $a = b + O_k$ for some $O_k \neq O_0$); and $[x] > [y] :\Leftrightarrow [x] \boxminus [y]$ is positive. This makes $\mathbb{Z}_{VR}$ a (decidably) ordered ring; positivity is decidable via the canonical form. (Order is inherited below by $\mathbb{Q}_{VR}$ and $\mathbb{R}_{VR}$; the symbols $\varepsilon > 0$, $|\cdot|$ used later refer to this order.)

II.6 Embedding $\mathbb{N} \rightarrow \mathbb{Z}_{VR}$. $n \mapsto [n \ominus 0]$.

Theorem 1. $\mathbb{Z}_{VR} = (\{[a \ominus b]\}, \oplus, \otimes, \boxminus, >)$ is an ordered commutative ring with unit, isomorphic to $\mathbb{Z}$.

Proof sketch. $\varphi([a \ominus b]) = a - b$ is well-defined on classes, respects operations and order, and is bijective.

Part III. Rationals $\mathbb{Q}$ as an operational superstructure

III.1 Language of division. A symbol $\oslash$; an expression $a \oslash b$ ($a, b \in \mathbb{Z}_{VR}$, $b \neq 0_{\mathbb{Z}}$) records “divide a by b ”. The restriction $b \neq 0_{\mathbb{Z}}$ is syntactic, decidable via the canonical form.

III.2 Equivalence. $a \odot b \approx_{\mathbb{Q}} c \odot d \Leftrightarrow a \otimes d = b \otimes c$ (integral-domain cancellation on $\mathbb{Z}_{VR}$; see §VIII.3).

III.3 Operations. $(a \odot b) \boxplus (c \odot d) := (a \otimes d \oplus b \otimes c) \odot (b \otimes d)$; $(a \odot b) \boxtimes (c \odot d) := (a \otimes c) \odot (b \otimes d)$; $(a \odot b) \boxminus_{\mathbb{Q}} (c \odot d) := (a \otimes d \boxminus b \otimes c) \odot (b \otimes d)$; $(a \odot b) \odot_{\mathbb{Q}} (c \odot d) := (a \otimes d) \odot (b \otimes c)$, provided $c \neq 0_{\mathbb{Z}}$.

III.4 Canonical form. Make the denominator positive (in the $\mathbb{Z}_{VR}$ -order, §II.5) and reduce to lowest terms.

III.5 Order and embedding. $a \odot b$ (denominator made positive) is positive iff a is positive in $\mathbb{Z}_{VR}$; this orders $\mathbb{Q}_{VR}$. Embedding $\mathbb{Z}_{VR} \rightarrow \mathbb{Q}_{VR}$: $a \mapsto [a \odot 1_{\mathbb{Z}}]$.

Theorem 2. $\mathbb{Q}_{VR}$ with $\boxplus, \boxtimes, \boxminus_{\mathbb{Q}}, \odot_{\mathbb{Q}}, >$ is an ordered field, isomorphic to $\mathbb{Q}$.

Part IV. Real numbers $\mathbb{R}$ via Cauchy sequences (computable modulus)

Real numbers require infinite processes, captured operationally as finite descriptions of them.

IV.1 Functions as operational rules. A function $a : \mathbb{N} \rightarrow \mathbb{Q}_{VR}$ is an operational rule: a finite description (algorithm) producing $a(n) \in \mathbb{Q}_{VR}$ for each n . Functions are not sets of ordered pairs but operational primitives. Each is captured by its finite description; the collection of such descriptions is operationally countable. Operationally, a is a partial recursive (Church–Turing, equivalently Type-2 computable) function (Weihrauch, 2000); the construction coincides with the standard development of computable reals via computable Cauchy sequences (Pour-El & Richards, 1989) — here not one approach among several, but the only one, dictated by the operational position.

IV.2 Fundamental (Cauchy) sequences, with a computable modulus. A function $a : \mathbb{N} \rightarrow \mathbb{Q}_{VR}$ is *fundamental* if

$$\forall \varepsilon \in \mathbb{Q}_{VR}, \varepsilon > 0, \exists N \in \mathbb{N} : \forall m, n \geq N : |a(m) \boxminus_{\mathbb{Q}} a(n)| < \varepsilon.$$

Crucially, the witness N must itself be *operational*: a **modulus of convergence** $N(\varepsilon)$ given by a finite description (an algorithm computing N from ε). This is the operational reading of $\exists N$ (§VI.3) and is not an extra hypothesis but the content of “operational rule”. Without a computable modulus a (computable) *Specker sequence* — classically Cauchy but with a non-computable limit — would qualify, and its class would fall outside the computable reals, breaking §IV.7. With the modulus computable, the limit is computable (Pour-El & Richards). The Lean realisation cannot express this restriction in a single type and proves the wider classical statement; see §VIII.6.

IV.3 Equivalence. $a \approx_C b \Leftrightarrow \forall \varepsilon > 0, \exists N : \forall n \geq N : |a(n) \boxminus_{\mathbb{Q}} b(n)| < \varepsilon$ (with a computable modulus, as in §IV.2).

IV.4 The construction. $\mathbb{R}_{VR}$ is the set of $\approx_C$ -classes of fundamental sequences.

IV.5 Operations. Componentwise on representatives: $[a] \oplus_{\mathbb{R}} [b] := [n \mapsto a(n) \boxplus b(n)]$; $[a] \otimes_{\mathbb{R}} [b] := [n \mapsto a(n) \boxtimes b(n)]$; $[a] \ominus_{\mathbb{R}} [b] := [n \mapsto a(n) \boxminus b(n)]$. *Division* $[a] \oslash_{\mathbb{R}} [b]$, for $[b] \neq 0$, requires a representative **bounded away from zero** (apartness): since $[b] \neq 0$ there are k and a positive rational δ with $|b(n)| > \delta$ for all $n \geq k$; choosing such a representative, $[a] \oslash_{\mathbb{R}} [b] := [n \mapsto a(n) \oslash_{\mathbb{Q}} b(n)]$ (for $n \geq k$) is well-defined. Apartness from 0, not mere $[b] \neq 0$, is what makes componentwise division total on a chosen representative.

IV.6 Embedding $\mathbb{Q} \rightarrow \mathbb{R}_{VR}$. $q \mapsto [n \mapsto q]$ (constant sequence).

IV.7 Isomorphism theorem.

Theorem 3. $\mathbb{R}_{VR}$ is a real closed ordered field, isomorphic to the field of computable real numbers — a countable subfield of the classical real line $\mathbb{R}$.

Remark 1. $\mathbb{R}_{VR}$ is not isomorphic to the entire classical $\mathbb{R}$: the latter has the cardinality of the continuum, whereas $\mathbb{R}_{VR}$, consisting of operational rules each given by a finite description with a computable modulus, is countable. Concretely, $\mathbb{R}_{VR}$ captures exactly the reals *admitting an algorithm that produces arbitrarily close rational approximations* (the computable reals) — the algebraic numbers, π , e , and so on. A real merely *definable* by a finite description need not be computable (e.g. Chaitin’s Ω , or the halting real) and is not among them; non-computable reals of classical $\mathbb{R}$ have no place here, corresponding to no algorithm. This aligns VR-Numbers with the constructive tradition (Bishop, 1967; Martin-Löf, 1984).

Remark 2 (relation to the continuum). This $\mathbb{R}_{VR}$ is the *Cauchy/computable* operational real — countable. It is a distinct construction from the choice-sequence (Brouwer) continuum pursued elsewhere in the cycle, which is non-enumerable; the two are different operational readings of “the reals”, not the same object.

IV.8 Operational character. A real in VR is, formally, an algorithm: a finite description specifying how to compute arbitrarily close rational approximations (e.g. $\sqrt{2}$ by Newton’s method).

Part V. Complex numbers $\mathbb{C}$ via the duality of A1

In standard mathematics $\mathbb{C}$ is built as ordered pairs of reals with a special multiplication, the two-dimensionality postulated through the choice of pairs. We ground it instead in the duality present in the generativity principle A1.

V.1 The duality of A1. A1 records two distinct generating facts: $F \rightarrow F = \top$ (a movement returning to F — self-reference at the logical level) and $F \rightarrow \top = \top$ (a movement to the opposite value $\top$). We read these as two independent axes of numerical extension.

V.2 The two axes. The axis $F \rightarrow F$ is the *real* axis ($\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R}$ unfolds along it). The axis $F \rightarrow \top$ is the *imaginary* axis, where expressions $b \cdot i$ live; i is not a number but a syntactic marker of which axis a value belongs to.

V.3 The joining rule. To make $\mathbb{C}$ one structure rather than two disjoint copies of $\mathbb{R}$, the axes are coupled by $i \otimes i := \ominus 1$ (i.e. -1): applying the imaginary marker twice equals negation along the real axis. This joining rule is an additional postulate, separate from A1’s duality.

V.4–V.7. A complex number is a formal expression $a \oplus b \cdot i$ ($a, b \in \mathbb{R}_{VR}$) — a syntactic record of two values along two axes, not an ordered pair. Equivalence: $a \oplus b \cdot i \approx c \oplus d \cdot i \Leftrightarrow a = c$ and $b = d$ in $\mathbb{R}_{VR}$. Operations: $(a \oplus b \cdot i) \oplus_{\mathbb{C}} (c \oplus d \cdot i) := (a \oplus_{\mathbb{R}} c) \oplus (b \oplus_{\mathbb{R}} d) \cdot i$; $(a \oplus b \cdot i) \otimes_{\mathbb{C}} (c \oplus d \cdot i) := (a \otimes_{\mathbb{R}} c \oplus_{\mathbb{R}} b \otimes_{\mathbb{R}} d) \oplus (a \otimes_{\mathbb{R}} d \oplus_{\mathbb{R}} b \otimes_{\mathbb{R}} c) \cdot i$; conjugation $a \oplus (\ominus b) \cdot i$; modulus $\sqrt{a \otimes_{\mathbb{R}} a \oplus_{\mathbb{R}} b \otimes_{\mathbb{R}} b}$. Embedding $\mathbb{R} \rightarrow \mathbb{C}_{VR}$: $a \mapsto a \oplus 0 \cdot i$.

Theorem 4. $\mathbb{C}_{VR}$ is a field, isomorphic to the standard $\mathbb{C}$ (with $\mathbb{R}$ replaced by $\mathbb{R}_{VR}$; the result is the field of computable complex numbers).

V.9 Foundation of two-dimensionality. A1 contains exactly two generating implications; these correspond to two distinct axes, so the two-dimensionality of $\mathbb{C}$ is not arbitrary but has a logical origin. *Scope:* A1’s duality gives two generative directions but does not by itself entail their algebraic coupling — the rule $i^2 = \ominus 1$ is an additional joining postulate. The claim is that the two-dimensionality is non-arbitrary, not that all of $\mathbb{C}$ ’s algebra is derivable from A1 alone. The theorems of Frobenius (1878) and Hurwitz (1898) (no further normed division algebra without weakening commutativity/associativity) are consistent with A1, as a logical primitive, not subdividing further; whether $\mathbb{H}, \mathbb{O}$ admit analogous derivations is open (Part VII).

Part VI. The operational position of VR-Numbers

VR-Numbers ascribes no ontology: there is no ontological primitive — $\emptyset$ is a nullary operation, characterised solely by $\forall x \neg(x \in \emptyset)$. All numbers are operational constructions (terms over the operations) of progressively greater depth.

VI.1 No ontological primitive. $\emptyset$ carries no ontological content — its entire characterisation is $\forall x \neg(x \in \emptyset)$, a fact about $\in$, not a description of contents. Nothing is “prior to operations”; $\emptyset$ is itself an operation. t is a rule, not an object; the $O_n = t^n(\emptyset)$ are records of successive applications, and that is the totality of what they are.

VI.2 Depths of operationality.

System	Operational character
$\emptyset$	The nullary base operation — depth zero of the construction (no ontology)
$\mathbb{N}$	Direct applications of t to $\emptyset$: $O_n = t^n(\emptyset)$
$\mathbb{Z}$	Finite syntactic expressions $a \ominus b$ over $\mathbb{N}$, on the $\mathbf{F} \rightarrow \mathbf{F}$ axis
$\mathbb{Q}$	Finite syntactic expressions $a \oslash b$ over $\mathbb{Z}$, on the $\mathbf{F} \rightarrow \mathbf{F}$ axis
$\mathbb{R}$	Algorithms (computable modulus) producing rational sequences, on the $\mathbf{F} \rightarrow \mathbf{F}$ axis
$\mathbb{C}$	Two-axis syntax over $\mathbf{F} \rightarrow \mathbf{F}$ and $\mathbf{F} \rightarrow \top$, joined by $i^2 = \ominus 1$

VI.3 Operational infinity. The induction principle A4 is conventionally read as postulating actual infinity — the completed totality of all naturals as an object. We reject this reading: A4 asserts no infinite object, but the operational principle that any property holding for $\emptyset$ and preserved under t holds for every result of iterated t . The quantifier “for all n ” is operational — universal applicability of an operation, not membership in a completed set. We call this *operational infinity*.

Operational and actual infinity **agree on arithmetical consequences**: via the equivalence $VR \Leftrightarrow$ first-order PA, every *arithmetical* theorem provable with actual infinity is provable here with operational infinity (and conversely). They differ in what they posit as existing: actual infinity

asserts an infinite totality as an object; operational infinity asserts only that an unbounded operation is available. *Beyond arithmetic the two diverge*: classical mathematics with actual infinity proves, e.g., the uncountability of $\mathbb{R}$ and the existence of non-computable reals — theorems with no counterpart here, since $\mathbb{R}_{VR}$ is countable (§IV.7). The equivalence is therefore at the level of arithmetic, not of all classical mathematics; the operational position does not reconstruct theorems essentially depending on completed uncountable totalities. This is a feature: it bypasses the Aristotelian/intuitionist objection to completed infinities without losing arithmetic.

All references to “infinite” objects ($\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$) are read operationally: none is an object; each is a domain of applicability for operations over $\emptyset$. The set-theoretic vocabulary is retained for conformity with standard practice, with the operational reading understood (the operational reconstruction of that vocabulary is the subject of VR-Sets).

VI.4 The gradient of operationality. Each successive system is a deeper operational construction over $\emptyset$: naturals (direct t -iteration), integers/rationals (finite syntax with equivalence), reals (finite descriptions of infinite processes, with computable modulus), complex (a second axis from A1, joined by $i^2 = \ominus 1$). The gradient mirrors the historical introduction of each extension and the conceptual difficulty that accompanied it.

VI.5 What does not exist as an object. Ordered pairs (replaced by syntactic juxtaposition $a \ominus b, a \otimes b, a \oplus b \cdot i$), negative numbers, fractions, reals, complex numbers — none is introduced as a primitive object; each is a notation for an operational class or expression. *Nothing is — all is doing.*

VI.6 Consistency relative to ZF. $\mathbb{Z}_{VR}, \mathbb{Q}_{VR}, \mathbb{C}_{VR}$ are isomorphic to their standard counterparts; the computable reals form a definable subfield of $\mathbb{R}$ in ZF (Pour-El & Richards, 1989) and $\mathbb{R}_{VR}$ is isomorphic to it. With the standard ZF-models for $\mathbb{Z}, \mathbb{Q}, \mathbb{C}$ and the consistency of VR relative to ZF, VR-Numbers is consistent relative to ZF (a metatheoretic corollary).

Part VII. Open questions

- (1) **Type theory.** The operational view of functions ($\mathbb{R}$) and the marker i ($\mathbb{C}$) suit Martin-Löf-style type theories; the Lean 4 formalisation (Part VIII) is a first step.
- (2) **Beyond $\mathbb{C}$.** Whether $\mathbb{H}, \mathbb{O}$ (Frobenius 1878, Hurwitz 1898) admit analogous derivations from deeper logical structure.
- (3) **VR-Sets.** The operational reconstruction of set theory — membership as “ x was used in the construction of y ”, unifying well-founded (ZFC) and non-well-founded (ZFA) modes — is now available (Brouwer edition, DOI 10.5281/zenodo.20592428).
- (4) **Computational realization.** Each level admits direct realization for exact-arithmetic computation.

Part VIII. Notes from the Lean 4 formalisation

Parts II–V are formalised in Lean 4 (Reznik, 2026, DOI 10.5281/zenodo.20352057; on Part I’s DOI 10.5281/zenodo.20324240), ≈ 3600 lines (Integers/Rationals/Reals/Complex), no `sorry`/`admit`, axiom dependencies documented. Eight observations, in three groups.

VIII.1 Hidden lemma (§II.2): cancellation of addition. “Direct verification” splits into unequal steps; transitivity needs right-cancellation $a + c = b + c \Rightarrow a = b$, proved by induction on c using successor injectivity (P4), not by T1–T4 rewriting.

VIII.2 Hidden lemma (§II.3): left distributivity. $(a + b) \times c = a \times c + b \times c$ is not in T1–T4 (commutativity of $\times$ is not assumed) and is proved by induction on the right argument; the formalisation makes this restraint visible.

VIII.3 Hidden lemma (§III.2): multiplicative cancellation. Transitivity of $\approx_{\mathbb{Q}}$ needs $xz = yz \Rightarrow x = y$ ($z \neq 0$); proved constructively via `Int.natAbs`, avoiding `mathlib`’s general `IntegralDomain` (which would pull `Classical.choice`). Integral-domain cancellation on $\mathbb{Z}$ is constructive; the choice-dependence is infrastructural, not intrinsic.

VIII.4 Boundary: Classical.choice between $\mathbb{Z}$ and $\mathbb{Q}$. $\mathbb{Z}_{VR}$ is fully constructive (`Quot.sound`, `propext` only). At $\mathbb{Q}_{VR}$, every theorem on $+$, $\times$ depends on `Classical.choice`, because Lean Core’s `Rat` carries a coprimality field whose normalisation goes through `Nat.gcd`. This is a feature of Lean’s `Rat` representation, not of the rationals (a non-normalising `Rat` would stay constructive); the boundary matches the deepening of operationality at the syntactic $\rightarrow$ canonical transition (§VI.4).

VIII.5 Boundary: the quotient becomes vacuous at $\mathbb{C}$. $\mathbb{Z}_{VR}, \mathbb{Q}_{VR}, \mathbb{R}_{VR}$ share the pattern record/equivalence/quotient. At $\mathbb{C}_{VR}$ the equivalence is componentwise equality (§V.5), coinciding with built-in equality on a two-field structure: the quotient is vacuous, $\mathbb{C}_{VR}$ is the first type realised without `Quotient`. The depth at $\mathbb{C}$ is carried by the two-dimensionality (A1, $i^2 = \ominus 1$), not by an equivalence.

VIII.6 Boundary: inexpressibility of computability on $\mathbb{R}_{VR}$. §IV.1 stipulates functions $\mathbb{N} \rightarrow \mathbb{Q}_{VR}$ are operational rules and §IV.7 concludes $\mathbb{R}_{VR} \cong$ the computable reals. But the Lean type $\mathbb{N} \rightarrow \mathbb{Q}_{VR}$ contains *every* total function of that signature (cardinality of the continuum), so the formalisation proves an isomorphism with the *full classical* $\mathbb{R}$, not the computable subfield: Lean 4 has no type-level predicate separating computable from non-computable functions, so the operational restriction (§IV.1–IV.2, including the computable modulus) is metatheoretic relative to Lean. Recommended reading within this cycle: the formalisation faithfully realises the Cauchy construction (§IV.1–IV.6) and additionally proves a structural isomorphism with classical $\mathbb{R}$; the restriction to computable reals (§IV.2, §IV.7) is preserved as metatheoretic commentary. VR-Forms provides the formal register in which the operational/non-operational relation is addressed directly.

VIII.7 Pattern: tactics cleaner than named lemmas. `mul_zero`, `mul_right_cancel0`, `linarith`, `linear_combination` carry latent `Classical.choice` via typeclass generality on concrete numeric types; `ring`, `omega`, manual identities discharge the same goals Classical-free. Prefer tactics to named algebraic lemmas for axiom-minimality on concrete types.

VIII.8 Pattern: change vs simp. At three points in $\mathbb{R}_{VR}$, `simp` drove goals into incompatible forms (`Rat.blt`; `abs_neg` ring-normalisation; `Real.mk_eq`); replacing it with `change` (definitional restatement, no reduction) kept syntactic control. The lighter tactic is preferable where it suffices.

Summary

VR-Numbers extends VR with operational superstructures yielding $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$. Three theses: (1) *No ontology* — $\emptyset$ is a nullary operation; all objects are terms over operations; no ordered pairs are primitive. (2) *Actual infinity is replaced by operational infinity* — induction is universal applicability of an operation, agreeing with actual infinity on arithmetical consequences (via $\text{VR} \leftrightarrow \text{PA}$) while positing no completed totality. (3) *The two-dimensionality of $\mathbb{C}$ is grounded in the duality of A1* — the two implications $\mathbf{F} \rightarrow \mathbf{F}, \mathbf{F} \rightarrow \mathbf{T}$ as real and imaginary axes, coupled by the additional rule $i^2 = \ominus 1$. $\mathbb{Z}_{VR}, \mathbb{Q}_{VR}, \mathbb{C}_{VR}$ are isomorphic to their standard counterparts; $\mathbb{R}_{VR}$ (Cauchy sequences with a computable modulus) is isomorphic to the computable reals, a countable subfield of $\mathbb{R}$. Consistency relative to ZF follows metatheoretically. The position is minimal: *nothing is — all is doing*.

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